

Open to opportunities

Geri Capo

Multidisciplinary Software & ML Engineer

I build working systems across AI/LLMs, robotics, data engineering, and visualization — and go deep enough in each to ship, evaluate, and verify, not just prototype.

📍 City, Country

 github.com/gericapo98

 github.com/0x000-byte

 Email

AI, LLMs & Applied GenAI

Building and evaluating LLM-powered systems — reasoning, agents, NLP, speech, and using generative models to drive real engineering pipelines.

LLMs

LangChain

Hugging Face

Ollama

Apple MLX

NLP

ASR / Whisper

modelica-genai-lab

MY PROJECT

Automated Modelica model generation, compilation & verification with GenAI.
Pipeline: prompt → .mo model → OpenModelica simulation → numeric comparison (RMSE/tolerance) → optional STL property checks → deviation report.

GenAI

Modelica

OMPython

FMU/FMPy

RTAMT (STL)

Verification

View on GitHub  (<https://github.com/gericapo98/modelica-genai-lab>)

langchain-rpa-simkit

MY PROJECT

LangChain toolkit that simulates RPA / agent workflows with typed (pydantic) chains, packaged with uv and covered by tests.

LangChain

Agents

Python

pydantic

View on GitHub  (<https://github.com/0x000-byte/langchain-rpa-simkit>)

intent-engine

MY PROJECT

Natural-language intent engine: candidate generation, NL parsing, and ranking to map free text onto structured intents.

NLP

Intent detection

Python

View on
GitHub

↗ (https://github.com/0x000-byte/intent-engine)

llms-lab

MY PROJECT

Experiments in LLM reasoning — DeepSeek-Math evaluation and rStar self-play reasoning tests.

LLMs

Reasoning

Evaluation

View on
GitHub

↗ (https://github.com/gericapo98/llms-lab)

asr

MY PROJECT

Whisper-based pipeline that transcribes lecture audio into aligned SRT / VTT / TSV transcripts.

ASR

Whisper

Python

Local project

Robotics & Autonomy

ROS 2 robotics — reproducible setup, simulation workspaces, and an active interest in navigation research.

ROS 2 (Rolling)

colcon

Simulation

Navigation

ros2-common-setup

MY PROJECT

Reusable ROS 2 Rolling environment setup, including bring-up on macOS.

ROS 2

DevOps

macOS

View on
GitHub

↗ (https://github.com/0x000-byte/ros2-common-setup)

ros2_rolling_ws

MY PROJECT

A ROS 2 Rolling colcon workspace (src / build / install) for experiments.

ROS 2

colcon

Workspace

Local project

robot_sim

MY PROJECT

A ROS 2 simulation playground for trying out robotics ideas.

ROS 2

Simulation

Local project

Data Engineering

Lakehouse-style data pipelines and embedded analytical engines.

DuckDB

Apache Iceberg

Docker Compose

Python (uv)

pyduck-iceberg

MY PROJECT

A DuckDB + Apache Iceberg lakehouse pipeline with a command-line interface and a fully Dockerized stack.

DuckDB

Apache Iceberg

Lakehouse

CLI

Docker

View on
GitHub

[↗ \(https://github.com/gericapo98/pyduck-iceberg\)](https://github.com/gericapo98/pyduck-iceberg)

Data Visualization

Turning data and abstract ideas into animations and plots.

Manim

Julia (Plots)

Matplotlib

SVG / motion

visualization-lab

MY PROJECT

Hybrid toolkit for dynamic animations with Manim (Python) and plots with Julia — built for explainers, demo videos, and interactive storytelling.

Manim

Julia

Animation

SVG

View on
GitHub



(<https://github.com/0x000-byte/visualization-lab>)

Systems, Security & Tooling

Reproducible environments, network-security tooling, and pragmatic CLIs that get a job done.

Docker

Linux / Ubuntu

cloud-init

Go

Git

Snort 3

Zeek

mangodex-client

MY PROJECT

A Go command-line client for the MangaDex API.

Go

CLI

REST API

View on
GitHub



(<https://github.com/gericapo98/mangadex-go-cli>)

kahoot-dl

MY PROJECT

Python tool to bulk-download all public Kahoot quizzes for a given creator.

Python

CLI

Scraping

Local project

ubuntu-sandbox

MY PROJECT

A reproducible Ubuntu development sandbox provisioned via cloud-init.

Docker

Linux

IoT

cloud-init

View on
GitHub



(<https://github.com/0x000-byte/ubuntu-sandbox>)

my-latex-templates

MY PROJECT

A curated, compile-tested collection of LaTeX templates (article, two-column paper, Beamer, CV, thesis, and more).

LaTeX

Templates

View on
GitHub



(<https://github.com/gericapo98/my-latex-files>)

Foundations & Explorations

Academic grounding and side explorations — the breadth behind the projects.

Clustering & EM

Recommender systems

mCRL2

Unity / C#

machine-learning

MY PROJECT

ML coursework and notebooks covering clustering, expectation-maximization, and supporting notes.

ML

Clustering

EM

Notebooks

Local project

resyst-study

MY PROJECT

A self-built web study app (quiz + materials) for a Reactive Systems course.

Web

JavaScript

Study tool

Local project

resys

MY PROJECT

Reactive-systems modeling with mCRL2 — process algebra and model checking.

mCRL2

Formal Methods

Model checking

Local project

MyUnityProject

MY PROJECT

A Unity / C# sandbox for interactive 3D and game-development experiments.

Unity

C#

Game Dev

Local project

Ecosystems I work with

Codebases and ecosystems I've studied, run, or contributed small fixes to — not authored.

LLM ECOSYSTEM

- Hugging Face Transformers
- Ollama
- Apple MLX (mlx-lm)

ROBOTICS RESEARCH

- arena2d — 2D RL navigation
- flying_adversarial_patch — adversarial ML on UAVs

NETWORK SECURITY

- Snort 3 (IDS)
- Zeek (network security monitor)

Skills

LANGUAGES

AI / ML

Python Go Julia C# JavaScript
Bash C/C++ LaTeX

LLMs LangChain
Hugging Face Transformers Ollama
Apple MLX NLP / intent parsing
ASR (Whisper) Clustering & EM
Recommender systems

DATA

DuckDB Apache Iceberg Pandas
NumPy Data pipelines uv packaging

SIMULATION & VERIFICATION

OpenModelica OMPython FMU / FMPy
RTAMT (STL) mCRL2

SYSTEMS & SECURITY

Docker Linux / Ubuntu cloud-init
macOS Git Snort 3 (IDS) Zeek (NSM)

ROBOTICS

ROS 2 (Rolling) colcon Simulation
Navigation (research)

VISUALIZATION

Manim Julia Plots Matplotlib
SVG / motion

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Geri Capo · built as a self-hosted static site · 2026